Introduction to Basic and Advanced Statistical Modelling

Fundamentals of statistics

Alexandre Courtiol

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Population and sample

Population and sample — what are statistics for?

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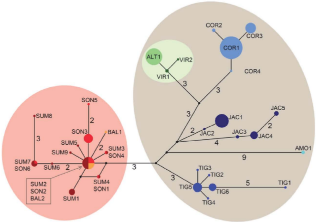

Haplotype network of tigers
(Wilting et al. 2015)

Population and sample — what are statistics for?

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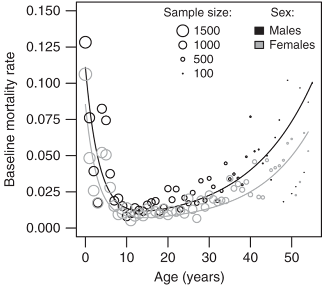

Yearly mortality rates of Myanmar timber elephants (Lahdenperä et al. 2018)

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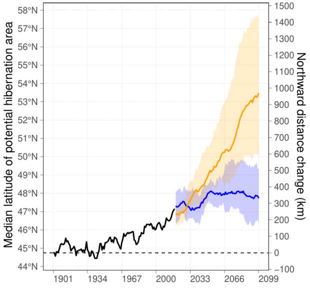

Potential hibernation area of common noctule bats (Kravchenko et al. 2025)

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Goals of statistics

- mediates between a *sample* and the *population*
- provides methods to draw as many conclusions as precisely as possible with as minimum *sampling effort* as possible

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Some practical questions we will tackle:

How large does the sample need to be? How to recognise and describe relationships among variables? How to quantify the reliability of the conclusions?

Population and sample — definitions

An adequate sample

a statistical sample should adequately represent the composition and properties of the entire population

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Is a blood sample an adequate statistical sample?

- Yes?

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- Yes?
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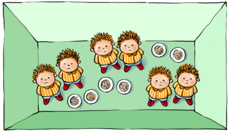
- It depends!

Population and sample — in practice

Which design would you prefer to study a new drug aiming at preventing stroke?

from Ulrich Dirnagl (Charité, Berlin)



DESIGN A

healthy, pubertal male twins raised in 6 m^2 isolator tents on an enriched granola diet

vs.

DESIGN B

patients of both sexes, elderly, comorbid, multiple medications, exposed to multiple pathogens and antigens throughout life

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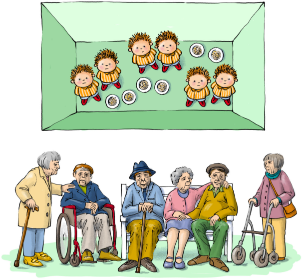

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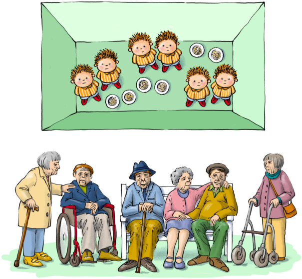

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The right *sampling design* may increase the external validity of your findings!

What would you mostly find in the medical literature?

Population and sample — external validity

RESEARCH ARTICLE | Originally Published 28 June 2016 |

Check for updates

Thomas Willis Lecture: Is Translational Stroke Research Broken, and if So, How Can We Fix It?

Ulrich Dirnagl, MD | AUTHOR INFO & AFFILIATIONS

Stroke • Volume 47, Number 8 • https://doi.org/10.1161/STROKEAHA.116.013244

"With few exceptions, stroke research is conducted in healthy, very young, male inbred rodent strains raised under specific pathogen-free conditions and fed a diet optimized for high fertility and overall health. The equivalent human cohort would be healthy pubertal twins raised in 6 m^2 isolator tents on an enriched granola diet, but patients with stroke are elderly, of both sexes, comorbid, on multiple medications, and have had exposure to numerous pathogens and antigens throughout their life. [...] As a general rule, the closer stroke models have mimicked patients with stroke [...], the more substantial has been the loss in efficacy of the treatment effects."

See also Voelkl, B., Würbel, H., Krzywinski, M. *et al.* The standardization fallacy. *Nat Methods* **18**, 5–7 (2021). https://doi.org/10.1038/s41592-020-01036-9

Population and sample — sampling effort

The *sampling effort* should:

- maximise the *sample size*
- minimise the *dependence* between objects
- (given constraints: time, money, impact...)

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Assuming that you only have a few minutes to sample hair in the classroom in order to estimate the average length of a piece of hair in the population, what should you do?

1. sample 10,000 pieces of hair on 2 individuals?
2. sample 1,000 pieces of hair on 10 individuals?
3. sample 500 pieces of hair on 4 males & 4 females?

Population and sample — sampling effort

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→ the best strategy depends on the extent to which hair length depends on sex & on the sex-ratio in *population*!

Population and sample — dependence between objects

Dependence between objects

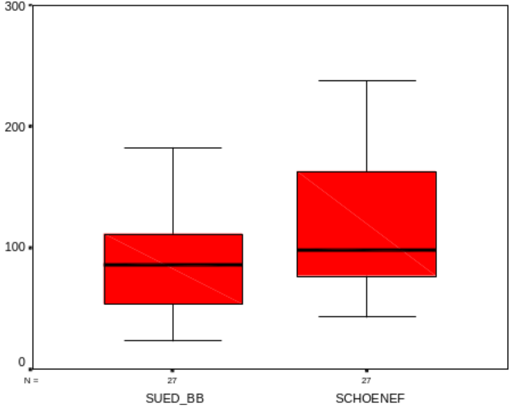
interrelationships between objects of the sample may lead to similarities of properties or measurements to be analysed

Usual sources of dependence

- time $\rightarrow$ generates *temporal autocorrelation*
- location $\rightarrow$ generates *spatial autocorrelation*
- measurements on the same individual $\rightarrow$ generates *pseudoreplication*
- pedigree/ancestry $\rightarrow$ generates *genetic correlation*

Population and sample — dependence between objects

Scenario I stress hormone concentrations in faeces of unknown roe deer, sampled in southern Brandenburg and in the proximity of an airport



Mann-Whitney U-test[1]:
$p = 0.079$; $n = 54$
difference of means: 26.5

[1] a non-parametric alternative to the unpaired t-test which is more robust

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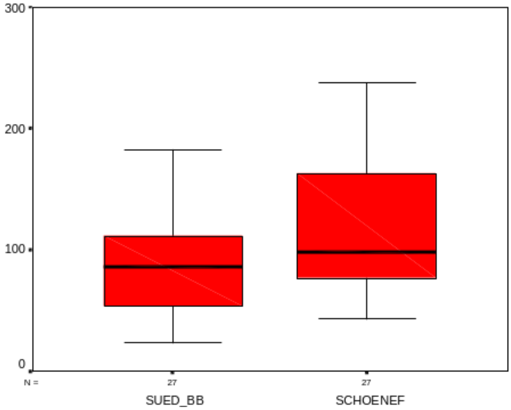


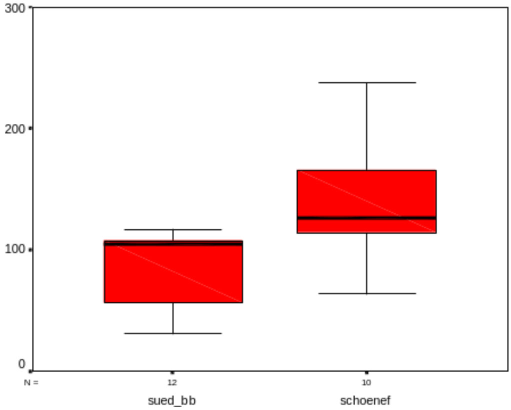
Mann-Whitney U-test[1]:
$p = 0.079$; $n = 54$
difference of means: 26.5

problem
population = faeces → *pseudoreplication!*

[1] a non-parametric alternative to the unpaired t-test which is more robust

Population and sample — dependence between objects

Scenario II stress hormone concentrations in faeces assigned to known roe deer individuals, sampled in southern Brandenburg and in the proximity of an airport



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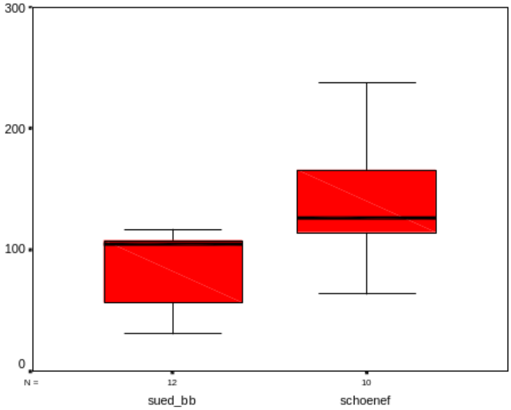

Mann-Whitney U-test[1]:
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solution

population = deer

→ fewer independent data points can contain more information than many dependent ones!

[1] a non-parametric alternative to the unpaired t-test which is more robust

Random variables and probability distributions

Random variables and probability distributions — definitions

- **random variable** a variable which can take different values

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Random variables and probability distributions — definitions

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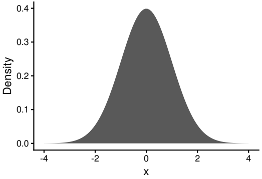
Because they are computed from a sample rather than from the entire population, the estimator is a *random variable* and its outcome is called the *estimate*!

Random variables and probability distributions — definitions

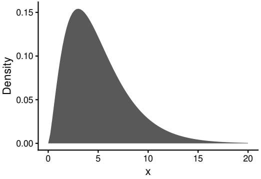
To describe the outcome of a *random variable*, we use *probability distributions*

Probability distributions

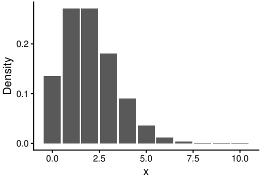
- are compact and comprehensive description of the *relative frequency* of all possible values
- comprises much more *information* than single estimator outcome
- are the most important tool of the theory for statistical estimation and *tests*
- serve as models for real data, which enables the estimation of *confidence intervals* and statistical *tests*



normal distribution



chi-squared distribution



Poisson distribution

Random variables and probability distributions — empirical description

Qualitative data

- observations (“measurements”) described by 2 or more *categories*
- special case: *binary* data (2 categories)

→ can be fully described by a *contingency table*, i.e. a table providing the number of observation for each category

Random variables and probability distributions — empirical description

Qualitative data

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→ can be fully described by a *contingency table*, i.e. a table providing the number of observation for each category

tidyverse

```
iris |> count(Species)
```

```
     Species  n
1     setosa 50
2 versicolor 50
3  virginica 50
```

base R

```
table(iris$Species)
```

```

    setosa versicolor  virginica 
        50         50         50 
```

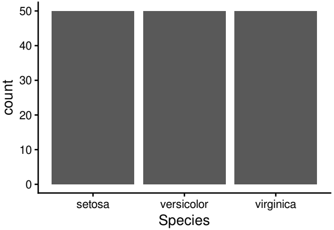
Random variables and probability distributions — empirical description

Qualitative data

- observations ("measurements") described by 2 or more *categories*
- special case: *binary* data (2 categories)

→ can be fully described by a *barplot* illustrating the contingency table

```
ggplot(iris) +
  aes(x = Species) +
  geom_bar()
```



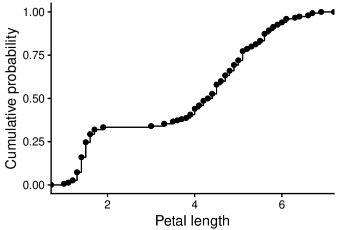
Random variables and probability distributions — empirical description

Quantitative data

- measurements (*continuous*) and counts (*discrete*)
- distance between each pair of data points is meaningful

$\rightarrow$ can be fully described by an *empirical cumulative distribution function* (ecdf)

```r
ggplot(iris) +
  aes(x = Petal.Length) +
  geom_step(stat = "ecdf") +
  geom_point(stat = "ecdf") +
  labs(y = "Cumulative probability",
       x = "Petal length")
```



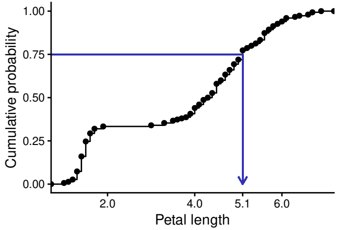
Random variables and probability distributions — empirical description

Quantiles

- the ecdf directly provides *quantiles*, i.e. the values at or below which given *proportions* of the data fall

```
quantile(iris$Petal.Length, probs = 0.75)
```

```
75%
5.1
```



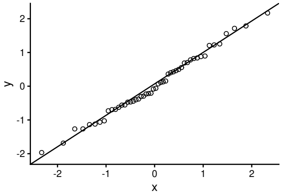
NB:

- reading up from a given cumulative probability (e.g. 0.75) to the curve, then across to the x-axis, identifies the corresponding *quantile* of petal length

Quantile–quantile plot

- the QQ plot compares 2 distributions by plotting their quantiles against each other
- it is often used to compare a distribution derived from the data to a normal distribution with same mean and variance (default in e.g. `geom_qq()`)

```
set.seed(123) # randomness is "fixed"
tibble(x = rnorm(n = 50)) |> # normal
ggplot(aes(sample = x)) +
  geom_qq(shape = 1) +
  geom_qq_line()
```



Random variables and probability distributions — comparing distributions

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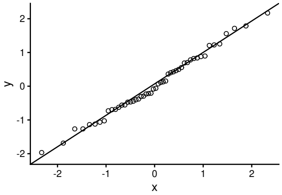

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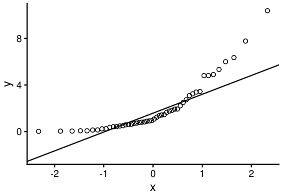
```
set.seed(123) # randomness is "fixed"
tibble(x = rchisq(n = 50, df = 2)) |> # chi-squared
ggplot(aes(sample = x)) +
  geom_qq(shape = 1) +
  geom_qq_line()
```

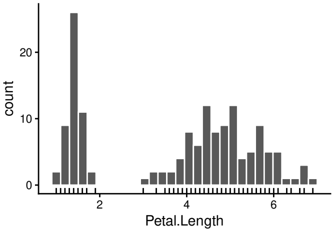


- if the points fall close to the QQ line, they are compatible with expectations under a *normal distribution*
- if many of the points fall far from the QQ line, they are not compatible with expectations under a *normal distribution* (see next course)

Quantitative data

a *histogram* does NOT fully describe a distribution

```
ggplot(iris) +
  aes(x = Petal.Length) +
  geom_histogram(colour = "white") +
  geom_rug()
```

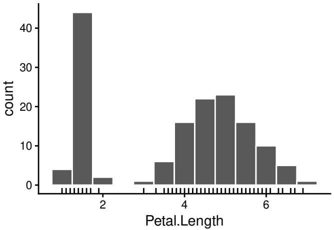


Random variables and probability distributions — comparing distributions

Quantitative data

a *histogram* does NOT fully describe a distribution

```
ggplot(iris) +
  aes(x = Petal.Length) +
  geom_histogram(colour = "white", binwidth = 0.5) +
  geom_rug()
```

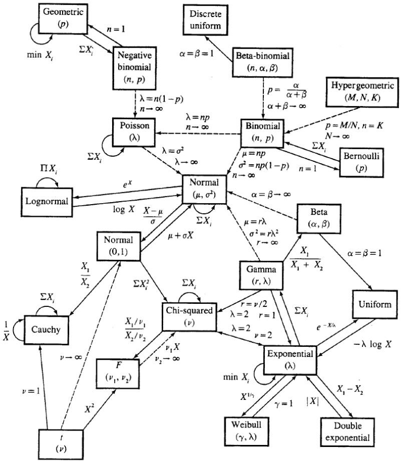


NB:

- the choice of bin width changes the perceived shape of the distribution
  $\rightarrow$ there is no single "correct" histogram for a given dataset
- not the best tool to check for normality (use Q-Q plots instead)

Random variables and probability distributions — theoretical distributions

630 *Table of Common Distributions*



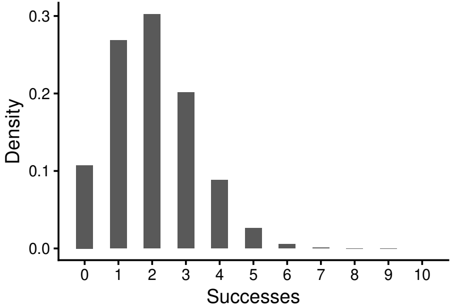
Relationships among common distributions. Solid lines represent transformations and special cases, dashed lines represent limits. Adapted from Leemis (1986).

from Casella & Berger *Statistical Inference*, 1990

Random variables and probability distributions — the binomial distribution

Probability density function (aka *pdf*) for a probability of success of 0.2, over 10 trials:

```
dbinom(x = 0:10, size = 10, prob = 0.2)
```



$$f(x,n,p) = \Pr(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

for $x = 0, 1, 2, \dots, n$, where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

The density gives the expected probability for exactly $\boldsymbol{x}$ successes in $\boldsymbol{n}$ *independent* binary trials if the probability for one success is equal to $\boldsymbol{p}$

Random variables and probability distributions — the binomial distribution

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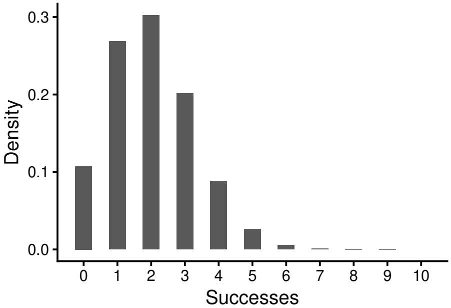

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for $x = 0, 1, 2, \ldots, n$, where

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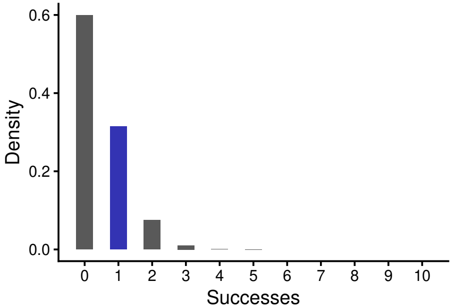
If the probability of catching a bird in one attempt is now expected to be 5%. Then, what is the chance that the biologist will capture exactly one bird in 10 attempts?

Random variables and probability distributions — the binomial distribution

Density for exactly one *success*, for a probability of success of 0.05 over 10 trials:

```
dbinom(x = 1, size = 10, prob = 0.05)
```



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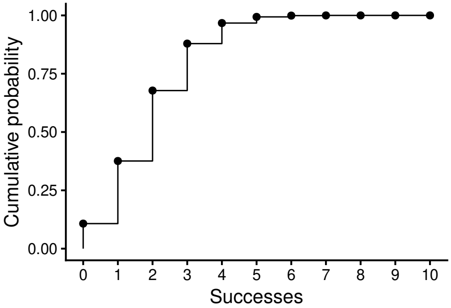
Random variables and probability distributions — the binomial distribution

Cumulative distribution function (*cdf*) for a probability of success of 0.2 over 10 trials:

```
pbinom(q = 0:10, size = 10, prob = 0.2)
```



$$F(x, n, p) = \Pr(X \leq x) = \sum_{i=0}^{\lfloor x \rfloor} \binom{n}{i} p^i (1-p)^{n-i}$$

where $$\binom{n}{i} = \frac{n!}{i!(n-i)!}$$

The cumulative probability gives the expected probability for ***x*** or fewer successes in ***n*** *independent* binary trials if the probability for one success is equal to ***p***

Random variables and probability distributions — the binomial distribution

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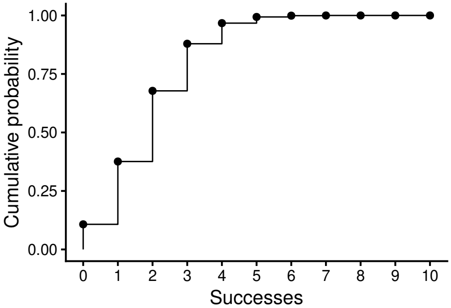

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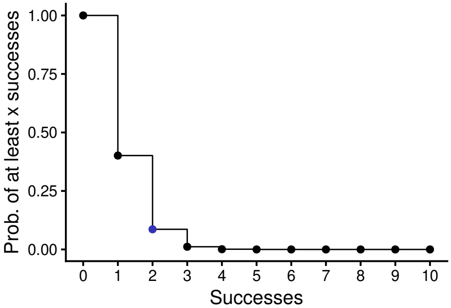
where $\binom{n}{i} = \frac{n!}{i!(n-i)!}$

If the probability of catching a bird in one attempt is expected to be 5%. Then, what is the chance that the biologist will capture at least two birds in 10 attempts?

Random variables and probability distributions — the binomial distribution

Probability of at least 2 successes for a probability of success of 0.05 over 10 trials:

```
1 - pbinom(q = 1, size = 10, prob = 0.05)
```



$$F(x,n,p) = \Pr(X \leq x) = \sum_{i=0}^{\lfloor x \rfloor} \binom{n}{i} p^i (1-p)^{n-i}$$

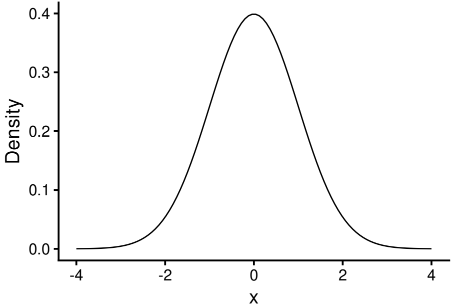
where $\binom{n}{i} = \frac{n!}{i!(n-i)!}$

If the probability of catching a bird in one attempt is expected to be 5%. Then, what is the chance that the biologist will capture at least two birds in 10 attempts?

Random variables and probability distributions — the normal distribution

Probability density function for a mean of 0 and a standard deviation of 1 (i.e. the standard normal distribution)

```
dnorm(x = seq(-4, 4, length.out = 100))
```



standard normal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

general normal distribution

$$f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

NB: it can be parametrised with mean (μ) and either the SD (σ) or the variance (σ^2), with $\widehat{\sigma^2} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \widehat{\mu})^2$ (note that the "$\widehat{\ }$" refers to estimates from data, as opposed to population values)

Random variables and probability distributions — the normal distribution

From general to standard: the *z-scores* transformation

if $X \sim \mathcal{N}(\mu, \sigma)$, then

$z = \frac{x - \hat{\mu}}{\hat{\sigma}} \sim \mathcal{N}(0, 1)$

Random variables and probability distributions — the normal distribution

From general to standard: the *z-scores* transformation

if $X \sim \mathcal{N}(\mu, \sigma)$, then

$z = \frac{x - \hat{\mu}}{\hat{\sigma}} \sim \mathcal{N}(0, 1)$

Back to general

if $Z \sim \mathcal{N}(0, 1)$, then given $\hat{\mu}$ and $\hat{\sigma}$, we can revert the z-scores transformation:

$x = (z \times \hat{\sigma}) + \hat{\mu}$

Random variables and probability distributions — the normal distribution

From general to standard: the *z-scores* transformation

if $X \sim \mathcal{N}(\mu, \sigma)$, then

$z = \frac{x - \hat{\mu}}{\hat{\sigma}} \sim \mathcal{N}(0, 1)$

Back to general

if $Z \sim \mathcal{N}(0, 1)$, then given $\hat{\mu}$ and $\hat{\sigma}$, we can revert the z-scores transformation:

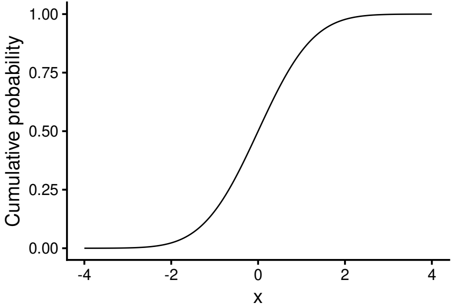
$x = (z \times \hat{\sigma}) + \hat{\mu}$

- the z-scores transformation is often used within statistical tests
- it can also help to understand how extreme some measurements are when units are not familiar (e.g. $z = 3$ correspond to a value 3 SD above the mean)
- outliers are often identified as observations with absolute z-scores $>$ 3 or 4, but very large outliers can inflate $\hat{\sigma}$ and thus be masked; in any case do not ditch outliers without discussing it!

Random variables and probability distributions — the normal distribution

Cumulative distribution function for a mean of 0 and a standard deviation of 1 (i.e. the standard normal distribution)

```
pnorm(q = seq(-4, 4, length.out = 100))
```



standard normal distribution

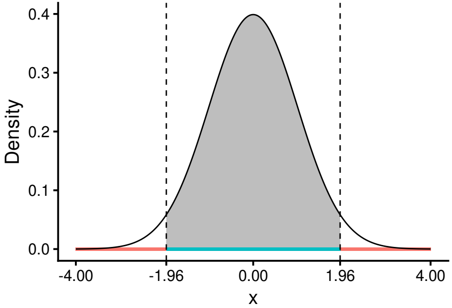
$$F(x) = \Pr(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \, dt$$

general normal distribution

$$F(x, \mu, \sigma) = \Pr(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} \, dt$$

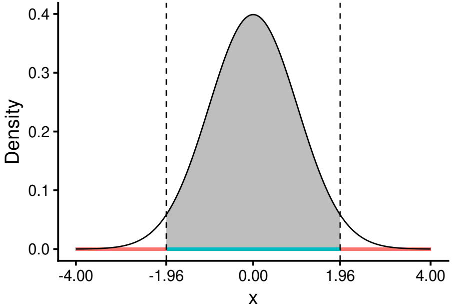
Random variables and probability distributions — the normal distribution

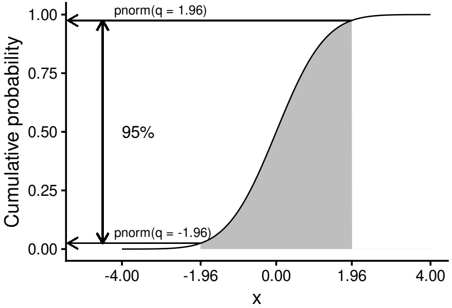
What is the expected probability that an outcome x will be between -1.96 and +1.96 if this event is the result of a standard normal distribution?



Random variables and probability distributions — the normal distribution

What is the expected probability that an outcome *x* will be between -1.96 and +1.96 if this event is the result of a standard normal distribution?





Random variables and probability distributions — the normal distribution

What is the probability for an observation following a normal distribution to have a value more extreme than:

- one SD?
- two SD?
- three SD?

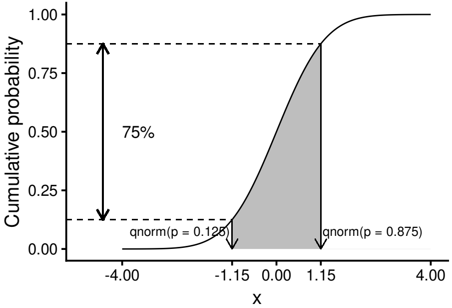
Instructions

- fix the mean to 0 and the SD to 1 to do the computation
- then, try doing the same with a different mean and SD

Reverted situation

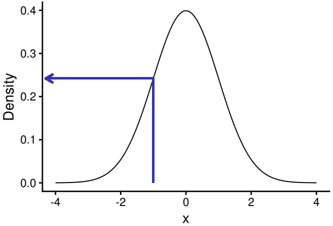
If random numbers are drawn from a standard normal distribution, where will the 75% of central observations fall?

$\rightarrow$ the answer is given by the *quantile function*



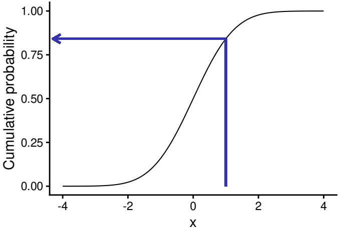
Random variables and probability distributions — summary

`dnorm(x = ...)`



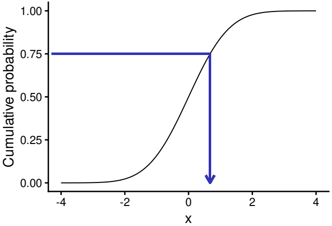
Probability density function

`pnorm(q = ...)`



Cumulative distribution function

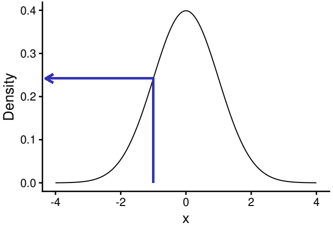
`qnorm(p = ...)`



Quantile function

Random variables and probability distributions — summary

`dnorm(x = ...)`



Probability density function

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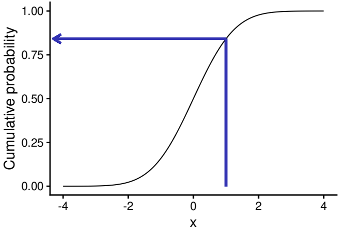
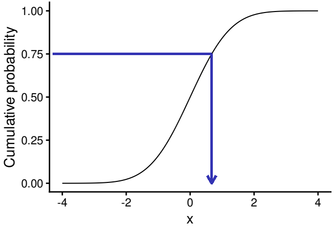


Cumulative distribution function

`qnorm(p = ...)`



Quantile function

NB:

- for the general normal distribution one needs to also define `mean = ...` and `sd = ...` in the function calls
- the same logic applies for all distributions (e.g. `dchisq(x = ..., df = ...)`, `dchisq(q = ..., df = ...)`, `qchisq(p = ..., df = ...)`)

Measuring uncertainty

Measuring uncertainty — estimation methods

NB: The distribution of the data is not the distribution of the estimator!

(even if one can be deduced from the other in some situations)

Estimation of uncertainty can be done:

- by assuming a distribution for the estimator
- by simulation (bootstrapping methods)

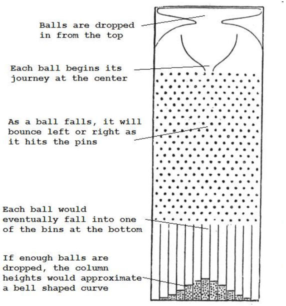
Measuring uncertainty — the central limit theorem

Central limit theorem

the mean of a sufficiently large number of *independently and identically distributed* random variables, each with finite mean and variance, will be approximately normally distributed

$\rightarrow$ we can assume that the estimator of the mean follows a *normal distribution*

Measuring uncertainty — the central limit theorem



Galton's box / quincunx / bean machine
Galton F (1889) *Natural inheritance*. Macmillan, London

NB: Francis Galton (1822–1911) was Charles Darwin's half-cousin (same grandfather: Erasmus Darwin).
He invented correlations, regressions, dog whistles but also "eugenics". He coined "nature vs nurture".
He also discovered anticyclones and was an explorer (e.g. Namibia).

Measuring uncertainty — metrics

Most widely used metrics

- the standard deviation of the estimator = the *standard error* (SE)

Example, the *standard error of the mean* (SEM):

$$\boxed{\text{SEM} = \frac{\text{SD}(\text{sample})}{\sqrt{n}}}$$

NB:

- SD quantifies how much measurements vary between each other
- SE quantifies how precise the estimation of a parameter is (i.e. how precise the estimation of the mean is if SE is a SEM)
- SEM varies with n (SD does not)

Measuring uncertainty — metrics

Most widely used metrics

- the standard deviation of the estimator = the *standard error* (SE)
- the 95% *confidence interval* ($\text{CI}_{95\%}$)
  $\rightarrow$ by definition, the procedure used for constructing the interval has to deliver a CI that includes the true value of the parameter in 95% of the time

Example: the *95% confidence interval for the mean*

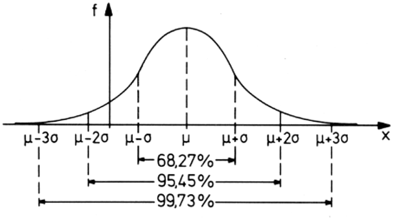
$\text{CI}_{95\%}$ for the mean
= mean $\pm$ qt(0.975, df)$\times$SEM
(df = degrees of freedom = the number of independent pieces of information available for the estimation = $n - 1$ if one mean estimated)
$\sim$ mean $\pm$ qnorm(0.975)$\times$SEM if n is large or SD(pop) known
$\sim$ mean $\pm$ 2$\times$SEM

Measuring uncertainty — metrics

Most widely used metrics

- the standard deviation of the estimator = the *standard error* (SE)
- the 95% *confidence interval* ($CI_{95\%}$)
  $\rightarrow$ by definition, the procedure used for constructing the interval has to deliver a CI that includes the true value of the parameter in 95% of the time

Example: the *95% confidence interval for the mean*



Likelihood

Likelihood — likelihood vs sampling distribution

Likelihood

the likelihood of a model given the data is equal to the *probability density* of those data given assumed parameter values and distribution of the data: $\mathcal{L}(\theta \mid y) = P(y \mid \theta)$

- the *sampling distribution* of the estimator shows how much estimates would vary if different samples were collected from the same population
- the *likelihood function* instead defines how plausible different estimates are given the sample you have

Likelihood — likelihood computation

Likelihood

the likelihood of a model given the data is equal to the *probability density* of those data given assumed parameter values and distribution of the data: $\mathcal{L}(\theta \mid y) = P(y \mid \theta)$

```
iris |>
  filter(Species == "versicolor") |>
  mutate(density = dnorm(x = Petal.Length, mean = 4.5, sd = 0.5),
         log_density = dnorm(x = Petal.Length, mean = 4.5, sd = 0.5, log = TRUE)) |>
  summarise(Lik = prod(density),
            logLik = sum(log_density))
```

```
           Lik    logLik
1 1.575195e-17 -38.68957
```

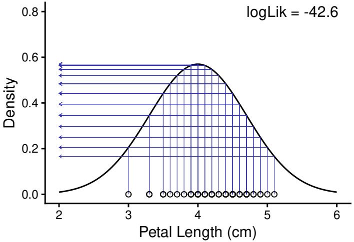
NB:

- using the log-likelihood produces values that are easier to work with, and avoids numerical *underflow* to zero (which the raw likelihood is prone to)

Likelihood — likelihood computation

Likelihood

the likelihood of a model given the data is equal to the *probability density* of those data given assumed parameter values and distribution of the data: $\mathcal{L}(\theta \mid y) = P(y \mid \theta)$



$\mu = 4$, $\sigma = 0.7$

Likelihood — likelihood computation

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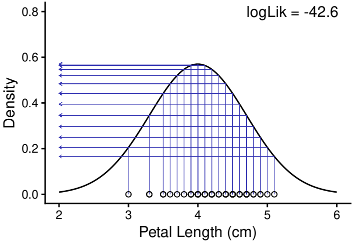
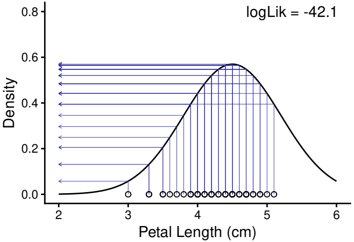


$\mu = 4,\ \sigma = 0.7$

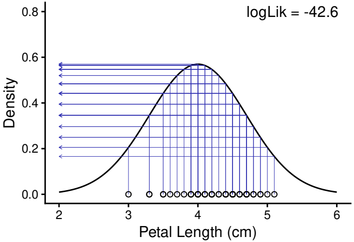


$\mu = 4.5,\ \sigma = 0.7$

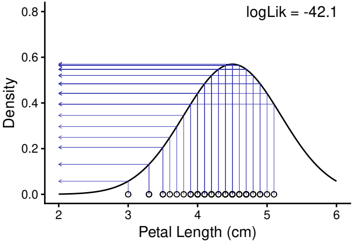
Likelihood — likelihood computation

Likelihood

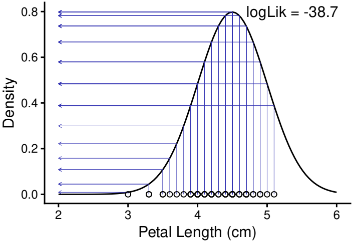
the likelihood of a model given the data is equal to the *probability density* of those data given assumed parameter values and distribution of the data: $\mathcal{L}(\theta \mid y) = P(y \mid \theta)$



$\mu = 4,\ \sigma = 0.7$



$\mu = 4.5,\ \sigma = 0.7$

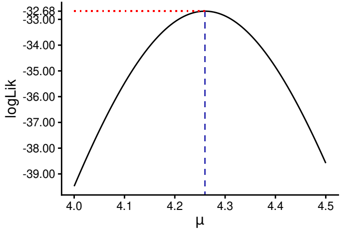


$\mu = 4.5,\ \sigma = 0.5$

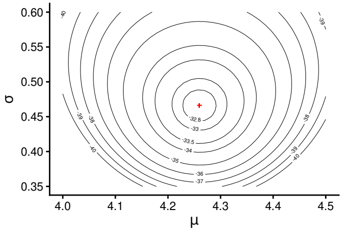
Likelihood — maximum likelihood estimates

Maximum likelihood estimates (MLE)

MLE are estimates that maximise the likelihood function



log-*likelihood profile* for μ

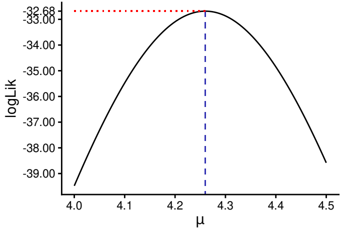


joint log-*likelihood surface* for μ & σ

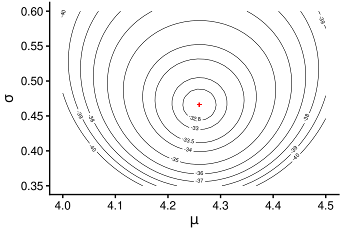
Likelihood — maximum likelihood estimates

Maximum likelihood estimates (MLE)

MLE are estimates that maximise the likelihood function



log-*likelihood profile* for μ



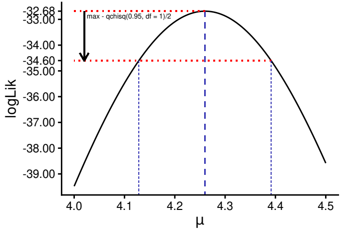
joint log-*likelihood surface* for μ & σ

- in this simple case the MLE of μ is the mean and the MLE of σ is the SD computed with an n-denominator rather than n−1, i.e. $\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i-\hat{\mu}\right)^2}$, or in R, `sd(x)*sqrt((n-1)/n)`

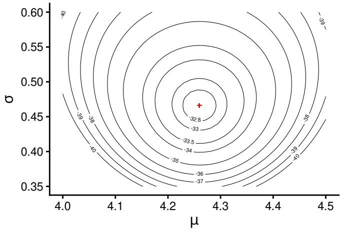
Likelihood — maximum likelihood estimates

Maximum likelihood estimates (MLE)

MLE are estimates that maximise the likelihood function



log-*likelihood profile* for μ



joint log-*likelihood surface* for μ & σ

- $\mathrm{CI}_{95\%}$ can also be derived from this approach

Tests and p-values

Tests and p-values — principles

Test setup

1. formulate a null hypothesis H0 for a population:
   $\rightarrow$ H0 is chosen in contrast to the experimental hypothesis (H1)
2. select and measure a sample
3. question: is the sample compatible with H0?
   - if "YES": by chance or reliably so?
   - if "NO": by chance or reliably so?

Tests and p-values — the null hypothesis

Ronald Fisher, *The Design of Experiments*, Edinburgh: Oliver and Boyd, 1935, p. 18

> *"In relation to any experiment we may speak of this hypothesis as the null hypothesis, and it should be noted that the null hypothesis is never proved or established, but is possibly disproved, in the course of experimentation.*
>
> *Every experiment may be said to exist only in order to give the facts a chance of disproving the null hypothesis."*

Tests and p-values — “testing by eye” using $\text{CI}_{95\%}$

- comparison of one estimated parameter $\hat{\mu}$ (e.g. a mean) with a fixed value μ_0 (H0: $\hat{\mu} = \mu_0$)
  - if μ_0 is located outside the $\text{CI}_{95\%}$, then $\hat{\mu}$ differs significantly from μ_0 (given $\alpha = 5\%$; more on α later)

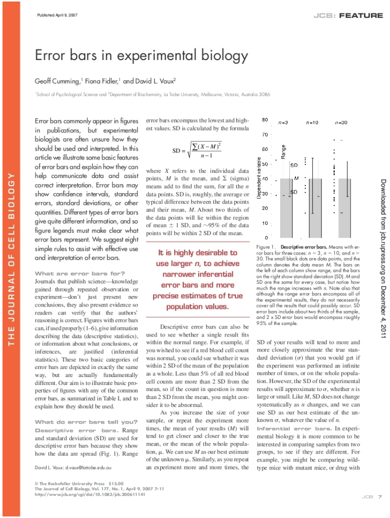
Tests and p-values — "testing by eye" using $\text{CI}_{95\%}$

- comparison of one estimated parameter $\hat{\mu}$ (e.g. a mean) with a fixed value μ_0 (H0: $\hat{\mu} = \mu_0$)
  - if μ_0 is located outside the $\text{CI}_{95\%}$, then $\hat{\mu}$ differs significantly from μ_0 (given $\alpha = 5\%$; more on α later)

- comparison of two estimated parameters $\hat{\mu}_1$ & $\hat{\mu}_2$ (e.g. two means) (H0: $\hat{\mu}_1 = \hat{\mu}_2$)

  a. if the two $\text{CI}_{95\%}$ do not overlap, then their means are significantly different
  b. if the two $\text{CI}_{95\%}$ overlap, then, mostly, the difference between the means is not significant; however, it may happen that, for small overlaps, the difference is significant $\rightarrow$ the $\text{CI}_{95\%}$ for the difference between two means $\neq$ the difference between two individual$\text{CI}_{95\%}$

Tests and p-values — “testing by eye” using $CI_{95\%}$



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JCB: FEATURE

Error bars in experimental biology

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Error bars commonly appear in figures in publications, but experimental biologists are often unsure how they should be used and interpreted. In this article we illustrate some basic features of error bars and explain how they can help communicate data and assist correct interpretation. Error bars may show confidence intervals, standard errors, standard deviations, or other quantities. Different types of error bars give quite different information, and so figure legends must make clear what error bars represent. We suggest eight simple rules to assist with effective use and interpretation of error bars.

What are error bars for?
Journals that publish science—knowledge gained through repeated observation or experiment—don't just present new conclusions, they also present evidence so readers can verify that the authors' reasoning is correct. Figures with error bars can, if used properly (1–6), give information describing the data (descriptive statistics), or information about what conclusions, or inferences, are justified (inferential statistics). These two basic categories of error bars are depicted in exactly the same way, but are actually fundamentally different. Our aim is to illustrate basic properties of figures with any of the common error bars, as summarized in Table I, and to explain how they should be used.

What do error bars tell you?
Descriptive error bars. Range and standard deviation (SD) are used for descriptive error bars because they show how the data are spread (Fig. 1). Range error bars encompass the lowest and highest values. SD is calculated by the formula

$$SD = \sqrt{\frac{\sum (X-M)^2}{n-1}}$$

where X refers to the individual data points, M is the mean, and Σ (sigma) means add to find the sum, for all the n data points. SD is, roughly, the average or typical difference between the data points and their mean, M. About two thirds of the data points will lie within the region of mean ± 1 SD, and ~95% of the data points will be within 2 SD of the mean.

It is highly desirable to use larger *n*, to achieve narrower inferential error bars and more precise estimates of true population values.

Descriptive error bars can also be used to see whether a single result fits within the normal range. For example, if you wished to see if a red blood cell count was normal, you could see whether it was within 2 SD of the mean of the population as a whole. Less than 5% of all red blood cell counts are more than 2 SD from the mean, so if the count in question is more than 2 SD from the mean, you might consider it to be abnormal.

As you increase the size of your sample, or repeat the experiment more times, the mean of your results (M) will tend to get closer and closer to the true mean, or the mean of the whole population, μ. We can use M as our best estimate of the unknown μ. Similarly, as you repeat an experiment more and more times, the SD of your results will tend to more and more closely approximate the true standard deviation (σ) that you would get if the experiment was performed an infinite number of times, or on the whole population. However, the SD of the experimental results will approximate to σ, whether n is large or small. Like M, SD does not change systematically as n changes, and we can use SD as our best estimate of the unknown σ, whatever the value of n.

Inferential error bars. In experimental biology it is more common to be interested in comparing samples from two groups, to see if they are different. For example, you might be comparing wild-type mice with mutant mice, or drug with

Figure 1. **Descriptive error bars.** Means with error bars for three cases: $n = 3$, $n = 10$, and $n = 30$. The small black dots are data points, and the column denotes the data mean M. The bars on the left of each column show range, and the bars on the right show standard deviation (SD). M and SD are the same for every case, but notice how much the range increases with n. Note also that although the range error bars encompass all of the experimental results, they do not necessarily cover all the results that could possibly occur. SD error bars include about two thirds of the sample, and 2 x SD error bars would encompass roughly 95% of the sample.

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The Journal of Cell Biology, Vol. 177, No. 1, April 9, 2007 7–11
http://www.jcb.org/cgi/doi/10.1083/jcb.200611141

Cumming et al. (2007) Error bars in experimental biology. *Journal of Cell Biology* 177(1): 7–11

Principles of statistical tests — test statistics

1. formulate a null hypothesis H0
2. define a test statistic, a function f suited to quantify the null hypothesis (e.g. difference of height means)
3. select a sample
4. hypothesise a distribution of f under H0; e.g. f is assumed to be normally distributed (bell-shaped)
5. according to H0, mean$(f) = 0$; SD(f) is derived from the selected sample
6. calculate the value $v = f$(sample) for the sample

Principles of statistical tests — test statistics

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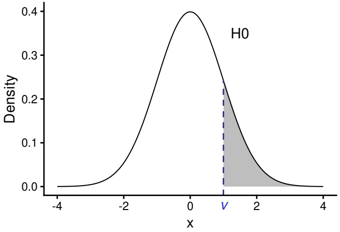
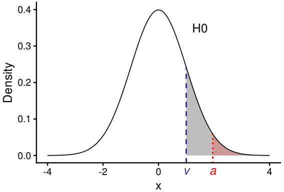
6. calculate the value $v = f$(sample) for the sample

Where is v located in the test distribution?
- v near the edge: **reject** H0
- v near the middle: **do not reject** H0



Principles of statistical tests — α and p-value



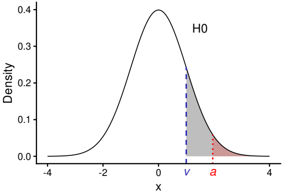
p-value

area beyond v (■), i.e. the probability to (randomly) obtain this sample or one with a more extreme departure from H0, if H0 is true

significance

p is regarded *significant* if p $< \alpha$ (■ $<$ ■), with α the *significance level*, i.e. the probability of rejecting the null hypothesis when it is true, previously defined by the researcher (traditionally: $\alpha = 0.05$)

Principles of statistical tests — α and p-value



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What is the significance outcome of the test illustrated above?

Principles of statistical tests — p-value

Data were collected on body height of males & females and the mean heights for males and females were calculated.

Result of a t-test: $t = 2.7$; $p = 0.01$

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Selection of suggested interpretations:

1. null hypothesis (H0: no difference) absolutely disproved
2. p-value reflects the probability of the H0
3. experimental hypothesis absolutely proved
4. p-value reflects the probability of the experimental hypothesis (different means)
5. p-value reflects the probability that the decision "reject H0" is wrong
6. for potential replications of the experiment, 99% of the occasions should be significant

Which ones are correct?

Principles of statistical tests — p-value

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6. for potential replications of the experiment, 99% of the occasions should be significant

Which ones are correct? None! $\rightarrow$ *frequentist statistics* cannot tell us the probability that a hypothesis is true; we will revisit this in the Bayesian section

Principles of statistical tests — significance

A difference between H0 and H1 is more easily detectable (“significant”)

- the larger the size (extent) of the biological effect is
- the smaller the variance of the measurements
- the larger the sample size is

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To evaluate a difference between H0 and H1 one has to consider:

- the *effect size*
- the p-value together with sample size & variance
  (which together shape the *variance of the test statistics*)

Principles of statistical tests — p-value

Data were collected on body height of males & females, the mean heights for males (M) and females (F) were calculated. All four results listed are possible:

M = 180	F = 175	p = 0.001
M = 180	F = 175	p = 0.100
M = 180.0	F = 179.9	p = 0.001
M = 175	F = 180	p = 0.049

Why?

Principles of statistical tests — p-value

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Why?

Potential reasons for unexpected results

- chance (sample not representative of the population)
- sample is systematically biased (volleyball team)
- sample too small or even very large
- improper test
- test requirements not met

Principles of statistical tests — p-value

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Why?

$\rightarrow$ The evaluation of a test result requires:

- p-value
- effect size (relevant effect?)
- sample (proper selection method chosen?)
- sample size
- test and whether its requirements were met

Principles of statistical tests — summary on p-values

ASA statement on statistical significance and p-values (Wasserstein 2016)

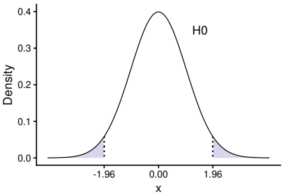
https://amstat.tandfonline.com/doi/pdf/10.1080/00031305.2016.1154108

- p-values can indicate how incompatible the data are with a specified statistical model
- p-values do not measure the probability that the studied hypothesis is true, or the probability that the data were produced by random chance alone
- scientific conclusions and business or policy decisions should not be based only on whether a p-value passes a specific threshold
- proper inference requires full reporting and transparency
- a p-value, or statistical significance, does not measure the size of an effect or the importance of a result
- by itself, a p-value does not provide a good measure of evidence regarding a model or hypothesis

Principles of statistical tests — one-sided vs two-sided tests

Two-sided hypothesis (e.g. body height): H0: males = females; H1: males $\neq$ females

- H0 is rejected only if the test statistic deviates in **either direction**
- "extreme" means that the test statistic in the 5%-region (i.e. the *rejection region*)
  $\rightarrow$ the "rejection region" is split into 2 $\times$ 2.5%-regions

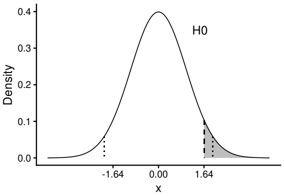


two-sided rejection regions (if the test statistic follows a standard normal distribution)

Principles of statistical tests — one-sided vs two-sided tests

One-sided hypothesis (e.g. body height): H0: males = females; H1: males > females

- H0 is rejected only if the test statistic deviates in **one direction**
- "extreme" means that the test statistic in the 5%-region (i.e. the *rejection region*)
  $\rightarrow$ the "rejection region" is at one side of the test distribution



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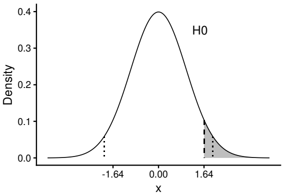

one-sided rejection region (if the test statistic follows a standard normal distribution)

One-sided hypotheses are easier to reject $\rightarrow$ use one-sided tests only if the opposite case can be very confidently excluded!

Principles of statistical tests — interpretation

How to interpret the result of null hypothesis significance testing (NHST)?
(e.g. the outcome of a t-test between the height of males and females)

- your test is significant $p \leq 0.05$
  $\rightarrow$ it supports that the (population) means of the two groups differ

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What is wrong with this classical interpretation of a lack of significance?

- the null hypothesis is dull
  $\rightarrow$ two groups most likely differ in reality, even if you don't have the power to detect the difference!

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What is wrong with this classical interpretation of a lack of significance?

Refined interpretation
$p > 0.05$ doesn't mean there is no effect
$\rightarrow$ your data are equally compatible with the effect going the other way
$\rightarrow$you cannot yet tell the direction of the true effect!

Multiple testing

Multiple testing — is alpha everything?

Question 1 if the null hypothesis is true in the population, what will be the probability to detect on a sample a significant departure from the null hypothesis (with $\alpha = 0.05$)?

$$\mathbf{P(reject\ H0 \mid H0) = ?}$$

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Question 2 if the null hypothesis is true in the population, what will be the probability to detect among 10 samples at least one showing a significant departure from the null hypothesis (with $\alpha = 0.05$)?

Cue: think about the biologist trying to catch a bird...

Multiple testing — is alpha everything?

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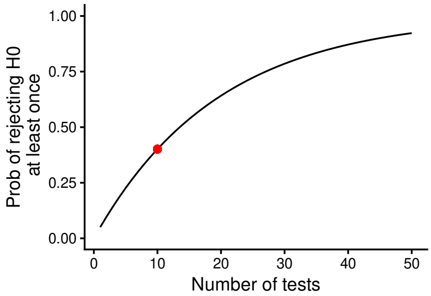
```r
1 - dbinom(x = 0, size = 10, prob = 0.05)
```

```
[1] 0.4012631
```



(H0 is assumed true)

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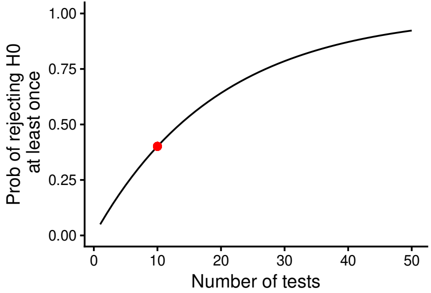
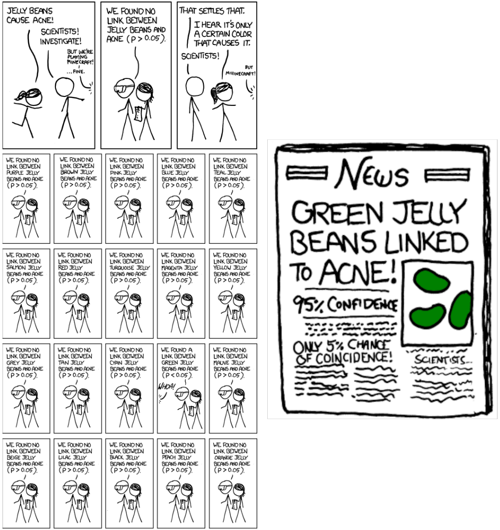


(H0 is assumed true)

How many experiments testing the same question are being conducted?

Multiple testing — is alpha everything?



© xkcd.com/882

Multiple testing — what to do?

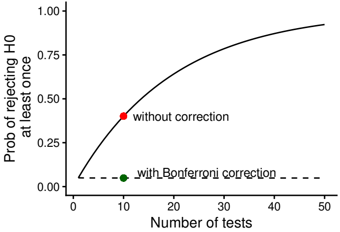
- avoid if possible!
- restrict the number of tests during the planning of analyses to “important” tests
- do not “generate” significant differences afterwards by excluding the non-significant tests
  $\rightarrow$ it is a recipe for generating false positives and thus a recipe for disasters (unethical & potentially harmful)
- there are special adjustment procedures if you must perform multiple testing

Multiple testing — the Bonferroni correction

Bonferroni correction

for m comparisons, test the single p-value not for $\alpha = 0.05$, but for $\alpha = 0.05/m$

Example: in the case of 10 tests, $\alpha = 0.05/10 = 0.005$



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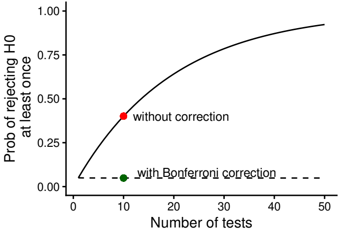

- the Bonferroni correction makes it much harder to detect real effects (we will define this cost, a decrease in *statistical power*, precisely soon)
- alternatives exist and differ in their trade-off between type 1 & 2 errors (see `?p.adjust`)

Multiple testing — replication vs multiple testing

Why should you replicate positive experiments (i.e. *cross validation*) but not negative ones (i.e. *multiple testing*)?

- if α is 0.05, what is the probability of obtaining at least one false positive after 3 experiments?

```r
1 - dbinom(x = 0, size = 3, prob = 0.05) # = 1 - 0.95^3
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- if α is 0.05, what is the probability of only obtaining false positives after 3 experiments?

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```

cross validation reduces the risk of false positive while multiple testing increases it!

Statistical power

Statistical power — type 1 and type 2 errors

prob. to given that	accept H0	reject H0
H0 true	true negative	false positive
H0 false	false negative	true positive

Statistical power — type 1 and type 2 errors

prob. to given that	accept H0	reject H0
H0 true	true negative	false positive
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Many metrics can be deduced from this table:

Metric	Definition
accuracy	true / all
sensitivity	true positive / H0 false
specificity	true negative / H0 true
precision	true positive / reject H0
false discovery rate	false positive / reject H0

Statistical power — type 1 and type 2 errors

prob. to given that	accept H0	reject H0
H0 true	true negative	false positive
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prob. to given that	accept H0	reject H0
H0 true	$1 - \alpha$	$\alpha =$ *type 1 error*
H0 false	$\beta =$ *type 2 error*	$1 - \beta =$ *power*

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H0 false	$\beta =$ *type 2 error*	$1 - \beta =$ *power*

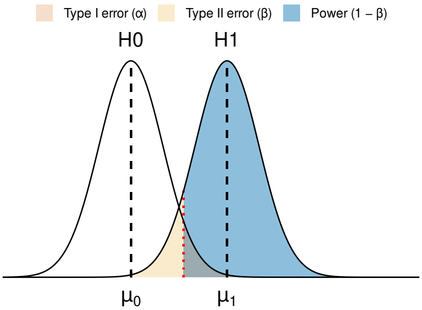
NB:

- if avoiding false negatives matters most, one should increase the power
- if avoiding false positives matters most, one should decrease α

... but there is a trade-off between the two, see next!

Statistical power — trade-off between false positives & negatives

- we can compute the distribution of the test statistic under H0
- let's also imagine that we know it under H1 (although it is not possible in practice)



baseline situation

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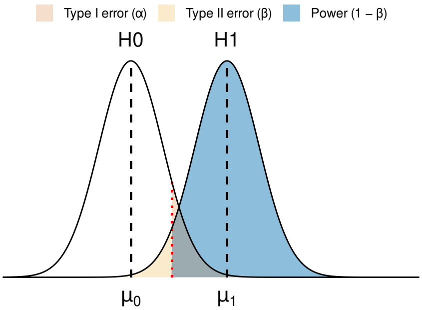

α increases $\rightarrow$ power increases but false positives increase too

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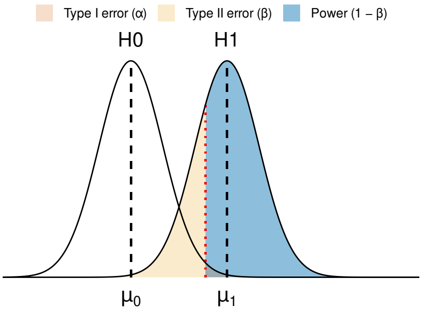

α decreases $\rightarrow$ false positives decrease but the power decreases too

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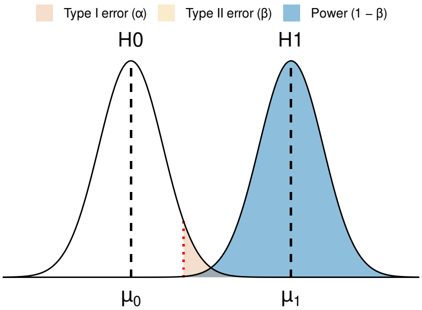

the trade-off decreases when the effect size is stronger

Statistical power — trade-off between false positives & negatives

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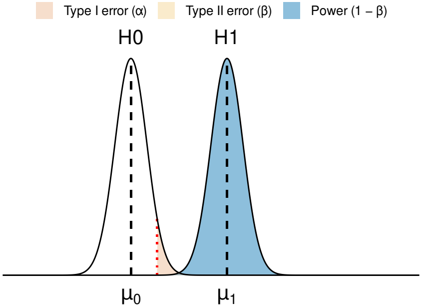

the trade-off decreases when the variances are smaller

Statistical power — planning of sample size

Ideal sample size?

- the power of any statistical test depends on sample size
- the sample should be large enough to detect a reasonable effect with an acceptably high probability

Why planning?

- sample size planning before data collection serves to prevent falsely insignificant results as a consequence of a sample being too small, and to prevent wasting time, effort and lives because a larger than necessary sample is being studied

Statistical power — planning of sample size

Interacting characteristics affecting a test result

1. power: common default: 0.8
2. α-risk: commonly 0.05
3. effect size: minimum relevant value
4. standard deviation: preliminary analysis
5. **sample size**: to be estimated

Statistical power — planning of sample size

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Frequent problems

- what is the minimum relevant effect?
- how to estimate the standard deviation (SD) without a preliminary analysis?

Statistical power — power analysis

Preparation

- get information about the (potential) SD of the sample
- define the size of the minimum relevant effect

Possible methods to plan the sample size

- use equations quantifying the relationship between α-risk, β-risk, effect size and SD for the test to be applied (e.g. `power.t.test()` for a t-test)
- simulate measurements and apply your test (works for all tests!)

Statistical power — power analysis by simulation

$\rightarrow$ for known α and n, the power can be determined by simulation, even if the distribution of the test statistic is unknown

Procedure

1. calculate the test statistic presuming H1 for many randomly generated samples of size n
2. determine the percentage of test results under H1, which are significant

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Procedure

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Constraint

you must have some idea concerning the process generating the data in order to generate the samples by an algorithm

Summary

What should you know before performing a statistical test?

- which question should be answered?
  more precisely: what is my null hypothesis?
- for which population is the question to be answered?
- which minimum effect size can be regarded as relevant?
- which is the right test to find the answer?
- which assumptions does the test statistic make about the population?
- which requirements apply to my sample in order
  1. to conform with the assumptions for the test statistic?
  2. to represent the population adequately?

Interlude

Interlude — is power everything?

Imagine an ideal situation: we perform a clinical test to diagnose a specific cancer with a known power of 99% and we set a significance threshold α at 0.05

Question 1 what is the probability of being positive at the test when having the cancer?

$$\mathbf{P(reject\ H0 \mid H1) = ?}$$

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$$\mathbf{P(reject\ H0 \mid H1) = ?}$$

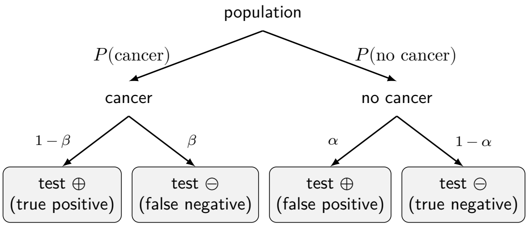
Question 2 what is the probability that a person with a positive test has the cancer?

$$\mathbf{P(H1 \mid reject\ H0) = ?}$$

NB: $\mathrm{P(H1 \mid reject\ H0)} \neq \mathrm{P(reject\ H0 \mid H1)}$

Interlude — is power everything?

Question 2 what is the probability that a person with a positive test has the cancer?



Interlude — is power everything?

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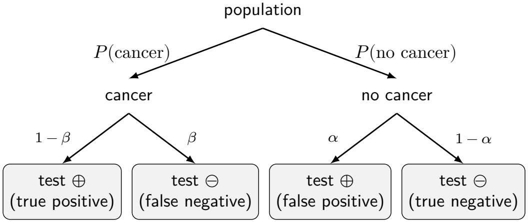

$$P(\text{cancer}|\oplus) = \frac{\text{true positive}}{\text{true positive}+\text{false positive}} = \frac{(1-\beta)\times P(\text{cancer})}{(1-\beta)\times P(\text{cancer})+\alpha\times P(\text{no cancer})}$$

- to chain probabilities (i.e. "and") you need to multiply them
- to combine probabilities (i.e. "or") you need to add them
- the vertical bar "|" means "conditional to" (i.e. $P(y|x)$ is the probability to observe y when x has occurred)

Interlude — is power everything?

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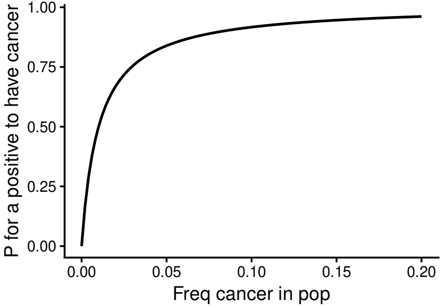
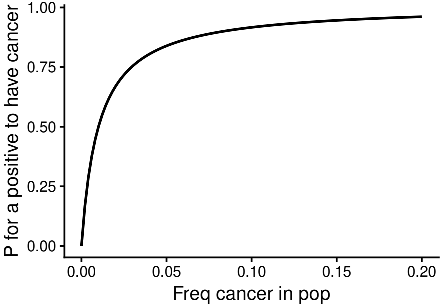


(computation assuming a power of 99% and α of 5%)

$\rightarrow$ it depends on the frequency of cancer in the population!

Interlude — is power everything?

Question 2 what is the probability that a person with a positive test has the cancer?



(computation assuming a power of 99% and α of 5%)

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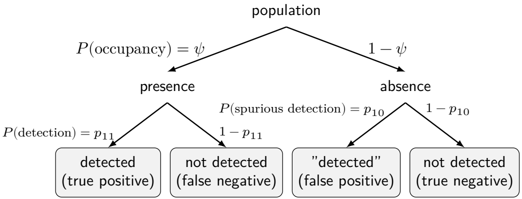
NB: you of all people should not be too surprised... think of camera traps!!

Interlude — a parallel with occupancy models

Question 3 what is the probability that a species not observed is not there?



Interlude — a parallel with occupancy models

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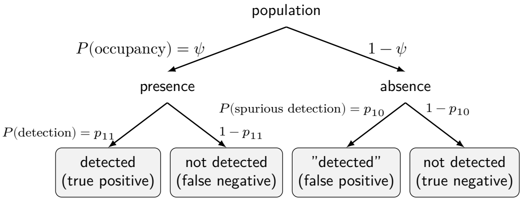


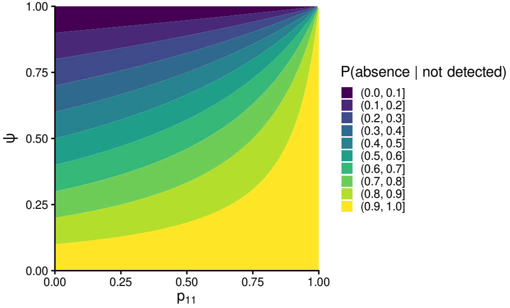
$$P(\text{absence}|\text{not detected}) = \frac{\text{true negative}}{\text{true negative}+\text{false negative}}$$

$$P(\text{absence}|\text{not detected}) = \frac{(1-p_{10})\times(1-\psi)}{(1-p_{10})\times(1-\psi)+(1-p_{11})\times\psi}$$

$$P(\text{absence}|\text{not detected}) \sim \frac{1\times(1-\psi)}{1\times(1-\psi)+(1-p_{11})\times\psi} = \frac{1-\psi}{1-p_{11}\psi} \text{ if } p_{10} \sim 0$$

Interlude — a parallel with occupancy models

Question 3 what is the probability that a species not observed is not there?



(computation assuming no spurious detection)

→ the probability of absence at a site where the species is not detected strongly depends on the probability of observation at sites where the species is present!

Bayesian statistics

Bayesian statistics — what are Bayesian statistics?

Bayesian statistics are an alternative branch of statistics where data analyses and parameter estimation are based on *Bayes' theorem*

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- large differences exist between alternative approaches:

	frequentist statistics	*Bayesian statistics*
probability interpretation	relative frequency of an outcome	degrees of belief
hypothesis status	either true or false	more or less probable
data representation	random draw from population	fixed
parameter representation	fixed value in population	probability distribution
external information	not formally incorporated	prior distributions
uncertainty measurement	confidence intervals	credible intervals

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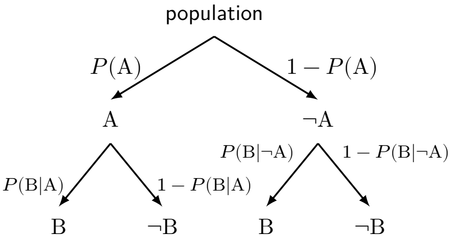
external information	not formally incorporated	prior distributions
uncertainty measurement	confidence intervals	credible intervals

- epistemological differences are interesting/intriguing but they are not the reason why Bayesian statistics have become increasingly popular in Ecology & Conservation
  $\rightarrow$ you will discover the practical reason soon enough

Bayesian statistics — Bayes' theorem

Abstract version of *Bayes' theorem*

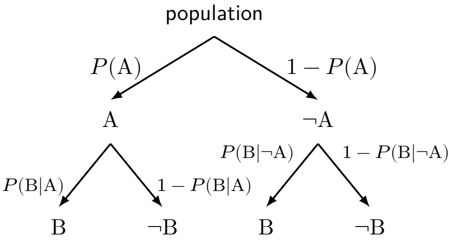


- $\neg\mathrm{X}$ means not X

$$P(\mathrm{A}|\mathrm{B}) = \frac{P(\mathrm{B}|\mathrm{A})\times P(A)}{P(\mathrm{B}|\mathrm{A})\times P(A)+P(\mathrm{B}|\neg\mathrm{A})\times P(\neg A)}$$

Bayesian statistics — Bayes' theorem

Abstract version of *Bayes' theorem*



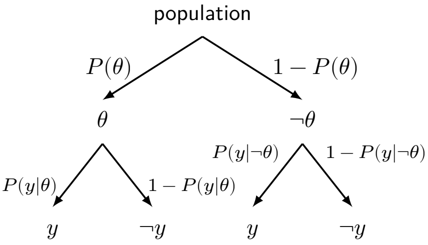
- $\neg$X means not X

$$P(\mathrm{A}|\mathrm{B}) = \frac{P(\mathrm{B}|\mathrm{A}) \times P(A)}{P(\mathrm{B}|\mathrm{A}) \times P(A) + P(\mathrm{B}|\neg\mathrm{A}) \times P(\neg A)}$$

$$\boxed{P(\mathrm{A}|\mathrm{B}) = \frac{P(\mathrm{B}|\mathrm{A}) \times P(A)}{P(\mathrm{B})}}$$

Bayesian statistics — Bayes' theorem

Relevant version of *Bayes' theorem*



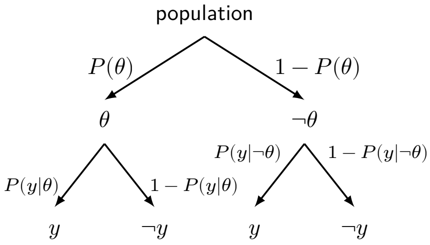
- $\propto$ means proportional
- θ = model parameters
- y = data

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

$$\boxed{P(\theta|y) \propto P(y|\theta) \times P(\theta)}$$

Bayesian statistics — Bayes' theorem

Relevant version of *Bayes' theorem*



- $\propto$ means proportional
- θ = model parameters
- y = data

$$P(\theta|y) = \frac{P(y|\theta) \times P(\theta)}{P(y)}$$

$$\boxed{P(\theta|y) \propto P(y|\theta) \times P(\theta)}$$

$\rightarrow$ the probability of the parameters given the data ($P(\theta|y)$), or *posterior*, is proportional to the product between the *likelihood* ($P(y|\theta)$) & the *prior distribution* ($P(\theta)$)

NB: *prior distribution* or *prior* represents the belief/uncertainty about the parameter value before observing the data

MCMC

- the Bayesian framework plays along well with *MCMC samplers*, a class of algorithms that can estimate distributions numerically
- given priors, MCMC samplers are used to approximate posterior distributions that are too complex or that have too many dimensions to be tackled analytically (the situation of many models used to study biodiversity)

Bayesian statistics — Markov chain Monte Carlo

MCMC

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Metropolis algorithm (1953)

- this is one of the simplest MCMC samplers
- it illustrates several key ideas related to Bayesian inference using MCMC
- it does not always perform well and modern software or packages use more modern (and more complex) MCMC samplers
- a well-known slightly more modern algorithm (1970) directly related to this sampler is called the *Metropolis-Hastings* algorithm

Bayesian statistics — Metropolis algorithm (principles)

Algorithmic steps:

1. define a prior distribution

Bayesian statistics — Metropolis algorithm (principles)

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2. initialise a *chain* (a vector) with a starting parameter value θ

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4. accept θ' if it increases the posterior density, otherwise accept it with a probability defined by the ratio of the posterior densities (i.e. $\frac{\mathrm{posterior}(\theta')}{\mathrm{posterior}(\theta)}$)

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7. drop the beginning of the chain (i.e. the *burn-in*) to remove the influence of starting values

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5. if θ' is accepted, add it to the chain, otherwise add θ again
6. repeat steps 3–5 until the chain reaches the target size
7. drop the beginning of the chain (i.e. the *burn-in*) to remove the influence of starting values
8. the remaining values in the chain are *samples from the posterior distribution*

Bayesian statistics — Metropolis algorithm (principles)

Key feature of the algorithm

- it does not require maximising the likelihood function; it only requires computing numerically the likelihood (or an approximation of it) at specific parameter values, which allows to fit much more complex statistical models

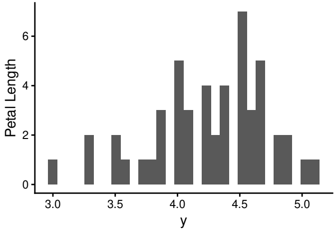
NB:

- the maximisation required in frequentist statistics to obtain point estimates is here replaced by a characterisation of the posterior distributions
- this is possible because the algorithm does not require computing the normalising constant of the posterior
  (the denominators of $\mathrm{posterior}(\theta')$ and $\mathrm{posterior}(\theta)$ cancel out in the ratio)
- this feature applies to many MCMC samplers and explain the success of Bayesian statistics in Ecology & Conservation

Bayesian statistics — Metropolis algorithm (application)

Let's use the Metropolis algorithm to estimate the mean and SD of petal length of *Iris versicolor*

```
iris |>
  filter(Species == "versicolor") |>
  pull(Petal.Length) -> y
```



distribution of the data
(not of the parameters!)

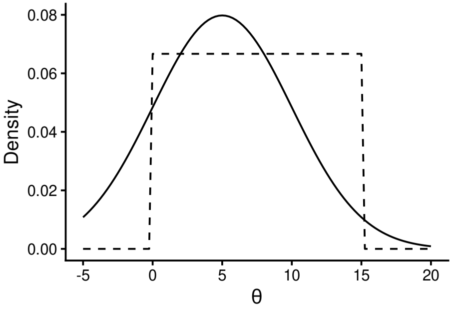
Bayesian statistics — Metropolis algorithm (homemade)

1. *prior elicitation*: we define prior distributions, here *weakly informative priors* (i.e. containing only crude information about the parameters)

```
my_prior_mu <- function(mu, log = TRUE) {
  # normal distribution
  dnorm(x = mu, mean = 5, sd = 5,
        log = log)
}

my_prior_sd <- function(sd, log = TRUE) {
  # uniform distribution
  dunif(x = sd, min = 0, max = 15,
        log = log)
}

my_logPrior <- function(param) {
  my_prior_mu(mu = param[1], log = TRUE) +
    my_prior_sd(sd = param[2], log = TRUE)
}
```



NB: priors can be multiplied (their log added) because we consider them independent, which simplifies things and is valid here

Bayesian statistics — Metropolis algorithm (homemade)

2. likelihood computation: we define a function to compute the log-likelihood

```
my_logLik <- function(param) {
  # make sure negative SD are not good
  if (param[2] < 0) return(-1e6)
  # compute the log-likelihood
  sum(dnorm(x = y,
            mean = param[1],
            sd = param[2],
            log = TRUE))
}
```

```
my_logLik(c(5, 0.5))
```

```
[1] -87.68957
```

```
my_logLik(c(5, 0.51))
```

```
[1] -85.71299
```

Bayesian statistics — Metropolis algorithm (homemade)

3. posterior computation: we define a function to compute the posterior values for each parameter

```r
my_logPosterior <- function(param) {
    # adding logs = multiplying originals
    my_logLik(param) + my_logPrior(param)
}
```

```r
my_logPosterior(c(5, 0.5))
```

```
[1] -92.92599
```

```r
my_logPosterior(c(5, 0.51))
```

```
[1] -90.94942
```

Bayesian statistics — Metropolis algorithm (homemade)

4. updating: we define a function that proposes new candidate parameter values close to the former ones

```r
my_proposal <- function(param) {
  rnorm(n = 2,
        mean = param,
        sd = c(0.1, 0.05))
}
```

```r
set.seed(123) # "fix" simu for reproducibility
my_proposal(param = c(5, 0.5))
```

```
[1] 4.9439524 0.4884911
```

```r
my_proposal(param = c(5, 0.5))
```

```
[1] 5.1558708 0.5035254
```

```r
my_proposal(param = c(5, 0.5))
```

```
[1] 5.0129288 0.5857532
```

Bayesian statistics — Metropolis algorithm (homemade)

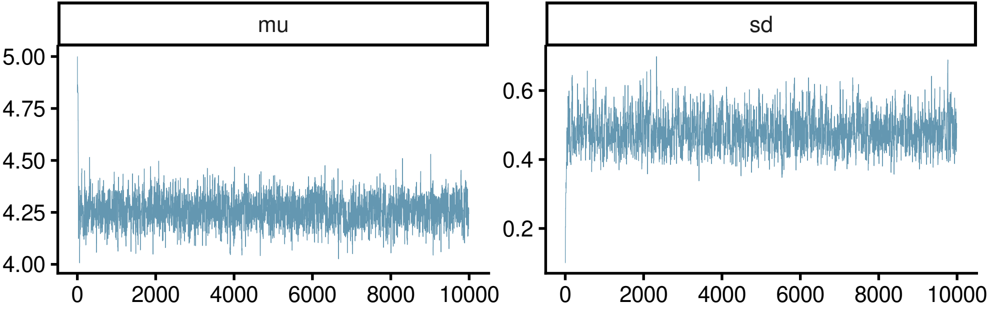
5. MCMC sampler: we build the Metropolis sampler

```
my_MCMC <- function(chain_length = 9999, starting_values = c(5, 0.1)) {
    chain <- matrix(nrow = chain_length + 1, ncol = 2) # for storing values
    chain[1, ] <- starting_values # add starting values
    for (i in seq_len(chain_length)) {
        param_candidate <- my_proposal(chain[i, ])
        proba_accept <- exp(my_logPosterior(param_candidate) -
                              my_logPosterior(chain[i, ])) # = ratio of posterior densitie
        if (runif(1) < proba_accept)
            chain[i + 1, ] <- param_candidate # new values accepted
        else
            chain[i + 1, ] <- chain[i, ] # former values retained
    }
    colnames(chain) <- c("mu", "sd") # name columns
    posterior::as_draws(chain) # convert matrix to "draws"
}
```

Bayesian statistics — Metropolis algorithm (homemade)

Let's run it!

```
library(posterior)
library(bayesplot)
posterior_samples <- my_MCMC()
bayesplot_theme_set(theme_classic(base_size = 30)) # set the default theme for plots
mcmc_trace(posterior_samples) # draw the chains
```

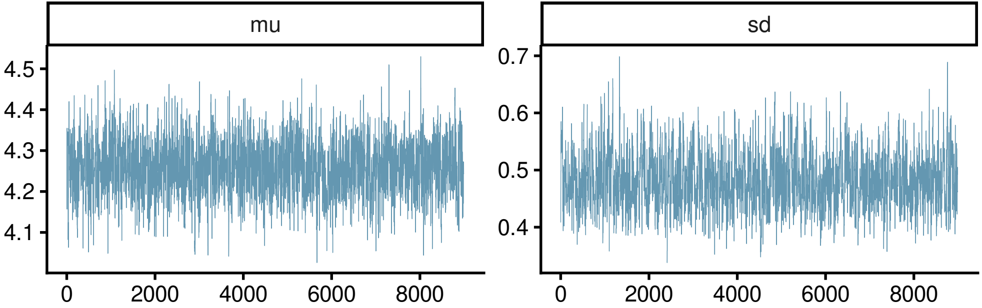


Bayesian statistics — clean up

Some clean up needs to occur:

1. discarding the *burn-in* part of the chain

```
burnin <- 1000
posterior_no_burnin <- posterior_samples[-seq_len(burnin), ] # subset (matrix like)
mcmc_trace(posterior_no_burnin) # trace are better -> more fuzzy caterpillar outlook
```

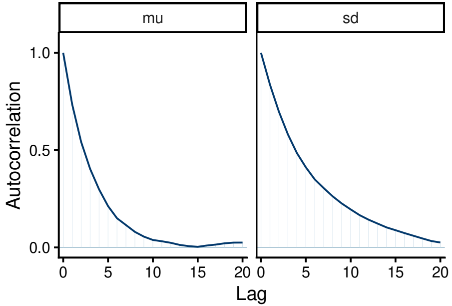


Bayesian statistics — clean up

Some clean up needs to occur:

1. discarding the *burn-in* part of the chain
2. *thinning* the chain (i.e. down-sampling to reduce autocorrelation)

```
mcmc_acf(posterior_no_burnin)
```

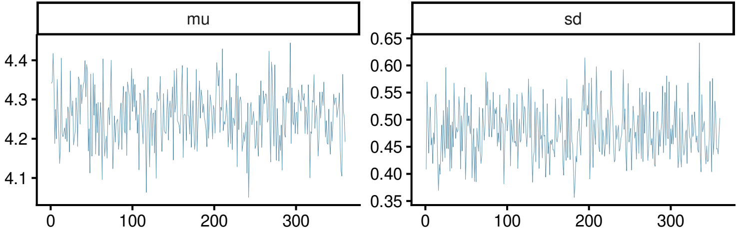



```
posterior_clean <- thin_draws(posterior_no_burnin, thin = 25)
```

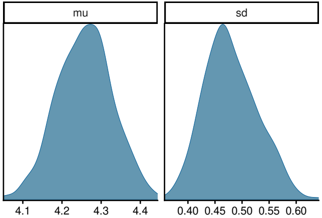
Bayesian statistics — results

Here are the results of our homemade posterior approximations:

```
mcmc_trace(posterior_clean)
```




```
mcmc_dens(posterior_clean)
```



Bayesian statistics — result checks

Then one would need to check the following:

- *acceptance rate* → to fine-tune the proposal function (not too high, not too low)
- *convergence of the chains* → use *trace plots* and diagnostics to ensure the algorithm reached the target distribution
- *effective sample size (ESS)* → to ensure you have enough independent samples after accounting for autocorrelation
- *sensitivity* to prior distribution → to make sure you did not introduce too much (or too little) information in them
- *posterior predictive checks* → to verify that the model can simulate data that looks like the real observations

NB:

- specific R packages exist to help you do that (e.g. {posterior}, {bayesplot})
- a good practice is to run several chains and compare them to make sure that all chains converge towards the same distribution (this is the basis of the Gelman-Rubin diagnostic)

Bayesian statistics — summarising the results

We can now describe the approximated posterior distributions using summary statistics:

```
summary(posterior_clean)
```

```
# A tibble: 2 x 10
  variable  mean median     sd    mad    q5   q95  rhat ess_bulk ess_tail
  <chr>    <dbl>  <dbl>  <dbl>  <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>
1 mu       4.26   4.26  0.0697 0.0673 4.15  4.37  0.999     239.     328.
2 sd       0.479  0.475 0.0489 0.0498 0.407 0.567 0.998     299.     301.
```

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2 sd       0.479  0.475 0.0489 0.0498 0.407 0.567 0.998     299.     301.
```

```
quantile2(posterior_clean, probs = c(0.025, 0.975)) # 95% credible interval
```

```
     q2.5     q97.5
0.4069504 4.3734805
```

Bayesian statistics — software

You won't have to do as much programming as we just did:

- for specific models, specific packages make it very easy to estimate posteriors (e.g. {brms}, {ubms})
- multi-purpose packages also makes things easier, although you would have to write the model part yourself (e.g. {nimble}, {rstan}, {rjags})

Bayesian statistics — nimble example

Let's revisit our example of estimating the mean and SD of petal length of *Iris versicolor* using this times {nimble}:

```
library(nimble)
## create code object
code <- nimbleCode({
  # defining model
  for (i in seq_len(N)) {
      y[i] ~ dnorm(mean = mu, sd = sigma)
  }
  # set priors
  mu ~ dnorm(mean = 5, sd = 5)
  sigma ~ dunif(min = 0, max = 15)
})

## define constants
constants <- list(N = length(y))
```

Bayesian statistics — nimble example

Let's revisit our example of estimating the mean and SD of petal length of *Iris versicolor* using this times {nimble}:

```r
## estimate parameters
results_nimb <- nimbleMCMC(code = code,
                           constants = constants,
                           data = list(y = y),
                           inits = list(mu = 5, sigma = 0.1),
                           nchains = 1, # 1 for trial, do more for real use (e.g 4)
                           progressBar = FALSE) # FALSE for not cluttering the slide
```

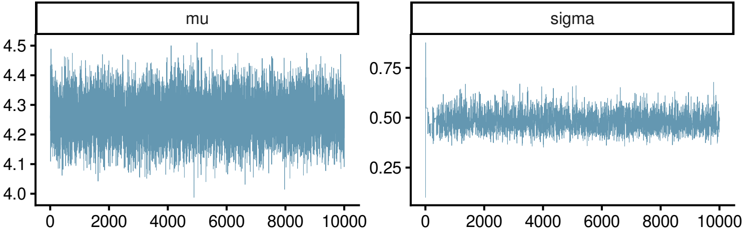
NB:

- {nimble} allows for much more complex workflow (e.g. you can define and configure the sampler you want to use)
- there are many companion packages extending {nimble} capabilities even further, e.g. {nimbleEcology} to define capture-recapture and occupancy models with ease

Bayesian statistics — nimble example

Then, you can work with outputs from {nimble} using {posterior} & {bayesplot}:

```
results_nimb_d <- as_draws(results_nimb)
mcmc_trace(results_nimb_d)
```



Bayesian statistics — Bayesian vs frequentist statistics

Both are useful

- sometimes you have no choice (a model can be too complex for most frequentist machinery)
- frequentist methods are generally faster and much easier to use (no prior elicitation needed)
- when the sample size is low, prior information can help the model converge faster and provides more stable estimates
- convergence is not guaranteed when using MCMC (must be verified via diagnostics)
- when the sample size is large, both should converge towards the same results since the influence of the prior decreases with n

Bayesian statistics — Bayesian vs frequentist statistics

Marc Kéry & Andy Royle, "Applied Hierarchical Modeling in Ecology"
eesc.usgs.gov/mbr/pubanalysis/keryroylebook/index.html

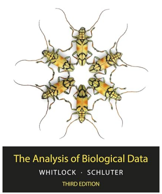
> *"Are all HMs Bayesian models? No, not at all! Although in practice HMs are perhaps more often fit using Bayesian than non-Bayesian methods, there is nothing intrinsically Bayesian about an HM (Cressie and Wikle, 2011). In our books we use both likelihood and Bayesian methods to fit HMs. Indeed, we are a little weary of what we perceive as completely unnecessary trench wars between anti-Bayesians and Bayesians … and we are very glad to note that they have ebbed down quite a bit during the last 2 decades. In such discussions, from the perspective of an applied statistical modeller, minute differences between the two approaches tend to be blown up completely out of proportion. We think that in practice the commonalities between non-Bayesian maximum likelihood and Bayesian inference are far, far greater than are their differences. Most of all, both likelihood and Bayesian inference are firmly rooted in the concept of a parametric statistical model, and in most practical applications the two will yield inferences that are numerically nearly indistinguishable. Of course, we do need to gloss over the major differences in the foundations of these methods… but we think that these are not something which will have to bother a non-statistician in most of his/her actual analyses. As a consequence, we firmly believe that both maximum likelihood and Bayesian methods are essential for applied statistical analysis, or applied data science, and the AHM project thoroughly reflects this view."*

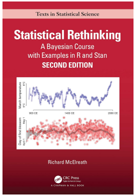
Learning more

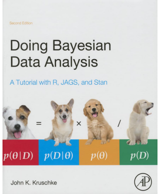
Learning more — some good reads

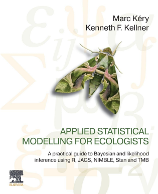
Useful reads

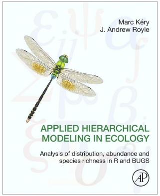
- www.nature.com/collections/qghhqm/pointsofsignificance
- van de Schoot et al. (2021) Bayesian statistics and modelling *Nat Rev Methods* doi.org/10.1038/s43586-020-00001-2

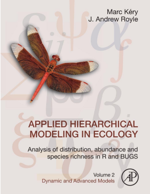












NB:

- all this material is unfortunately behind expensive pay-walls
- even in the age of LLMs, getting solid basics in statistics will make you more efficient, critical & confident

Learning more — statistics and the quest for knowledge

- watch the excellent video from Veritasium which summarises much of the frequentist content from this course
  www.youtube.com/watch?v=42QuXLucH3Q

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- accept the fact that much of the research findings is false, but contribute to build better science with your newly acquired or refreshed statistical knowledge


Open access, freely available online

Essay

Why Most Published Research Findings Are False

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It can be proven that most claimed research findings are false.

Ioannidis (2005) *PLoS Med* 2(8): e124

Questions?

Developer corner

Ignore this slide — (technical info for Alex)

```
[1] "2.6 min of R running time"

R version 4.6.1 (2026-06-24)
Platform: x86_64-redhat-linux-gnu
Running under: Fedora Linux 44 (KDE Plasma Desktop Edition)

Matrix products: default
BLAS/LAPACK: FlexiBLAS OPENBLAS-OPENMP;  LAPACK version 3.12.1

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

other attached packages:
 [1] nimble_1.4.2       bayesplot_1.15.0   posterior_1.7.0    geomtextpath_0.2.0 lubridate_1.9.5    forcats_1.0.1
 [7] stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2        readr_2.2.0        tidyr_1.3.2        tibble_3.3.1
[13] ggplot2_4.0.3      tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] tensorA_0.36.2.1     gtable_0.3.6         xfun_0.60            lattice_0.23-1       tzdb_0.5.0           numDeriv_2016.8-
1.1
 [7] CoprManager_0.5.8    vctrs_0.7.3          tools_4.6.1          Rdpack_2.6.6         generics_0.1.4       parallel_4.6.1
[13] proxy_0.4-29         pkgconfig_2.0.3      Matrix_1.7-6         checkmate_2.3.4      RColorBrewer_1.1-3   S7_0.2.2
[19] distributional_0.8.1 lifecycle_1.0.5      compiler_4.6.1       farver_2.1.2         textshaping_1.0.5    tinytex_0.60
[25] codetools_0.2-20     htmltools_0.5.9      yaml_2.3.12          pracma_2.4.6         nloptr_2.2.1         pillar_1.11.1
[31] MASS_7.3-66          reformulas_0.4.4     abind_1.4-8          boot_1.3-32          spaMM_4.6.65         nlme_3.1-170
[37] tidyselect_1.2.1     digest_0.6.39        stringi_1.8.9        slam_0.1-56          reshape2_1.4.5       labeling_0.4.3
[43] fastmap_1.2.0        grid_4.6.1           cli_3.6.6            magrittr_2.0.5       utf8_1.2.6           withr_3.0.3
[49] scales_1.4.0         backports_1.5.1      registry_0.5-1       timechange_0.4.0     rmarkdown_2.31       matrixStats_1.5.0
[55] igraph_2.3.3         otel_0.2.0           hms_1.1.4            coda_0.19-4.1        ROI_1.0-2            pbapply_1.7-4
[61] evaluate_1.0.5       knitr_1.51           rbibutils_2.4.1      viridisLite_0.4.3    rlang_1.3.0          isoband_0.3.0
[67] Rcpp_1.1.2           glue_1.8.1           rstudioapi_0.19.0    minqa_1.2.8          jsonlite_2.0.0       plyr_1.8.9
[73] R6_2.6.1             systemfonts_1.3.2
```